

Slot 5 · The format race

File: `format_bench.py` · **Done when:** `uv run pytest tests/test_slot05.py` shows 3 passed · **Time:** 12 min

Today's idea

The same data is saved in four ways: **CSV**, **Parquet**, **Parquet without compression** and **SQLite**. We ask all of them the same question and time each one.

The question is not only "which is fastest?". The real question is: **what did we give up to get that speed?**

The question we ask every time:

Note Average `eta_error_min` by city, for orders over 5 km. (Answer: 6 rows.)

Step 1 · Write a fair timer

Open `format_bench.py`. Write `timed(fn, repeats=3)`:

- run `fn()` three times
- measure each run with `time.perf_counter()`
- return **(the fastest time in milliseconds, the result of fn)**

Why the fastest of three? The first run is often slow because the file is not yet in memory. Then you are timing your disk, not the format.

Step 2 · Add the two missing racers

Four racers are already written in `race()`. Add two more, following the same pattern:

1. **DuckDB** on `data/orders_uncompressed.parquet` (copy the DuckDB line, change the file name)
2. **SQLite** on `data/orders.db` — the table is called `orders`:

```
con = sqlite3.connect("data/orders.db")
add("SQLite", "data/orders.db", lambda: con.execute(
    "select city, avg(eta_error_min) from orders "
    "where distance_km > 5 group by city").fetchall())
```

Step 3 · Race

```
uv run python format_bench.py
```

Your numbers will be different, but the **order** should be similar:

approach	file_MB	ms	x_fastest
DuckDB over uncompressed Parquet	93.3	2.4	1.0
DuckDB over Parquet	20.1	4.4	1.8
polars read Parquet	20.1	5.2	2.2
polars read CSV	90.7	24.1	10.0
SQLite	108.0	151.4	63.1
pandas read CSV	90.7	276.7	115.3

Tell the instructor your fastest and slowest numbers.

Step 4 · What did we give up?

Discuss with your neighbour and fill this in:

format	what is good	what we give up
CSV	anyone can open it, even Excel	
Parquet	small and fast	
Parquet, no compression	fastest here	
SQLite		

Think Uncompressed Parquet was fastest but **4.6× bigger**. Would it still win if the file was on a slow network drive instead of your SSD?

Step 5 · Check you are done

```
uv run pytest tests/test_slot05.py
```

If something goes wrong

what you see	what to do
<code>test_timed_takes_the_best_not_the_first</code> fails	You must call <code>fn()</code> exactly <code>repeats</code> times.
first run much slower than the rest	Normal. That is why we take the best of three.
<code>no such table: orders</code>	Run <code>make data</code> .
SQLite is the fastest	Check that you used the same question, not <code>count(*)</code> .